

AI Stock Price Predictor - Execution Report

Project Execution Details:

Date: 2026-05-16 16:04:55

Repository: AI-Stock-Price-Predictor

Environment: Python Backend (Flask) & React Frontend (Vite)

Modules Implemented & Integrated:

1. Backend API (Flask)

- Stock price fetching & technical analysis (RSI, SMA, EMA, MACD).
- XGBoost / ML Model integration for 30-day price prediction.
- XAI (Explainable AI) components providing actionable trading insights.

2. Frontend UI (React + Vite)

- Dynamic Dashboard for real-time stock insights.
- Interactive Recharts-based candlestick and line charts for historical & predicted prices.
- Explainer Sidebar rendering model confidences and technical signals objectively.

Execution Steps:

Step 1: Backend Setup

> cd Stock-Prediction-Dashboard-AI

> pip install -r requirements.txt

> python app.py

[SUCCESS] Flask API running on http://127.0.0.1:5000

Step 2: Frontend Setup

> cd Stock-Prediction-Dashboard-AI/frontend

> npm install

> npm run dev

[SUCCESS] React Vite dev server running on http://localhost:5173

AI Stock Price Predictor - Execution Report

Test execution confirmed backend-frontend connectivity.

Real-time predictions and charts are fully functional.

All code modules successfully staged and pushed to GitHub with descriptive commits.